SRIDHARAN N

PRODUCT DEVELOPMENT ENGINEER

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PROFILE

Product Development Engineer with **2.5 years of experience** in automotive new product development, specialized in **Electric motors development**, supporting end-to-end lifecycle from concept to validation and production release. Strong exposure to CAD modelling, GD&T, tolerance stack-up, DFM/DFA. Experienced in component development, prototype development, design verification (DVP), product validation support. Actively collaborating with simulation, manufacturing, quality, and suppliers to deliver robust, manufacturable, and validated products.

CORE SKILLS

Product Development: Design, drawing release, design documentation, New Product development, Prototyping

CAD & Design Tools: Creo, SolidWorks, NX CAD (basic)

Drafting & GD&T: ASME Y14.5, tolerance stack-up, interface & clearance validation, BOM (Bill of Materials)

Motors: BLDC, PMSM, SnyRM motors

Validation & Testing: Design verification (DVP), product validation support (PV), proto testing

Analysis: Engineering calculations, basic FEA, interpretation of simulation results

Quality Tools: DFMEA, 8D, Why-Why, Fishbone, VA/VE

Standards & Process: basic exposure on IATF 16949, APQP, PPAP, product lifecycle management

Tools: SAP (ERP), basic Windchill, PLM systems, MS Office

General: Supplier coordination, cross-functional collaboration, problem-solving

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PROFESSIONAL EXPERIENCE

SONA COMSTAR at Chennai, Tamil Nadu

Product Development Engineer, July 2024 to Present

- Contributed to end-to-end product development of electric motors from design to manufacturing release
- Translated customer and system requirements into fully constrained 3D CAD models and assemblies
- Performed tolerance stack-up analysis to ensure functional fit, assembly performance, and part interchangeability
- Executed engineering calculations and supported simulation result interpretation for design optimization
- Collaborated with cross-functional teams for material selection considering performance, cost, and manufacturability
- Created detailed 2D drawings with GD&T, fits, tolerances, and inspection requirements as per standards
- Developed and maintained BOMs and ensured accuracy of engineering documentation
- Participated in DFMEA to identify potential failure modes and improve product reliability

- Conducted DFM/DFA studies at part, sub-assembly, and system levels to enhance manufacturability and assembly efficiency
- Supported prototype builds, trials, and iterative design improvements based on testing and manufacturing feedback
- Coordinated with suppliers for component development, drawing clarification, and CTQ alignment
- Built and validated proof-of-concept (POC), prototypes and implemented design improvements from validation feedback
- Conducted validation tests including electrical, performance testings
- Assisted in DVP execution and product validation activities including test planning, execution, and reporting
- Worked closely with simulation, manufacturing, and quality teams to resolve design and production issues
- Ensured design data integrity and version control across PLM and manufacturing systems
- Performed root cause analysis (RCA) and 8D problem-solving for development and field issues

Project Exposure

- Worked on electric motor development PMSM (Permanent Magnet Synchronous motor), BLDC (Brushless DC motor), SRM (Synchronous Reluctance motor) for EV applications
- Customers – Royal Enfield, TVS motor, Mahindra, TI.

INTERNSHIPS

Product Development Engineer Intern at **SONA COMSTAR**, Chennai, Tamil Nadu
Feb 2024 – Jun 2024

- Design and development of Electric motors.

Graduate Innovation Engineer Trainee at **FORGE INNOVATION AND VENTURES**, Salem, Tamil Nadu
Feb 2023 – Jun 2023

- Designed and developed functional prototypes integrating mechanical, electromechanical, and sensor-based systems, utilised additive manufacturing techniques for Rapid prototyping.
- Supported prototype testing, data collection, and analysis using Python and Excel.
- Documented design concepts, test observations, and improvement recommendations.

EDUCATION

B.E. Mechanical Engineering — Government College of Engineering, Salem | 2020 – 2024 | CGPA: 9.37/10

HSSC — South Indian Matriculation Hr. Sec. School | 2020 | 83.6%
SSLC — South Indian Matriculation Hr. Sec. School | 2018 | 95%
